Insurance Fraud Detection Using Machine Learning
Project Report

1. Project Title

Insurance Fraud Detection Using Machine Learning

2. Project Description

Insurance fraud is a serious problem that causes significant financial losses to insurance companies every year.

This project presents a machine learning based system that can analyse insurance claim data and detect whether a claim is fraudulent or genuine.

The system uses historical claim data and classification algorithms to identify suspicious patterns.

The project also provides a simple user interface built using Streamlit so that users can input claim details and receive fraud prediction results.

3. Application Scenarios

Enterprise Usage: Insurance companies can integrate the system to automatically detect suspicious claims before approving them.

End-user Usage: Fraud investigation teams can use the model to analyse claim details.

Educational Usage: Students can study how machine learning is used for fraud detection.

Industrial Usage: Financial organizations can adapt the system for banking or transaction fraud detection.

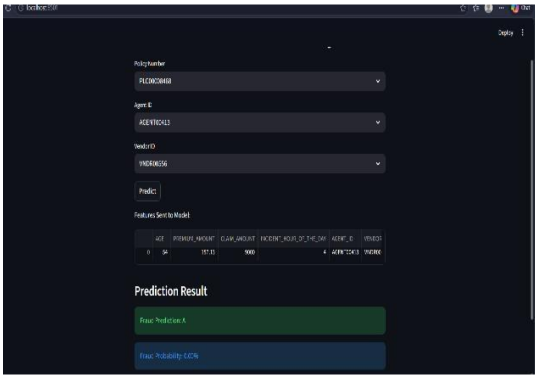
4. Technical Architecture Overview

The system architecture contains three main layers:

1. User Interface Layer – Built using Streamlit where users provide claim inputs.
2. Processing Layer – Handles preprocessing and data transformation.
3. Machine Learning Layer – Predicts fraud using trained models.

Data Flow:

User Input → Data Processing → ML Model Prediction → Output Display.



5. Prerequisites

Software Requirements:

Python 3.x

Streamlit

VS Code / PyCharm

Libraries:

Pandas

NumPy

Scikit-learn

Joblib

Matplotlib

Hardware Requirements:

Processor: Intel i3 or higher

RAM: Minimum 4GB

Storage: At least 500MB free space.

6. Prior Knowledge Required

Basic understanding of:

Python programming

Machine learning concepts

Data preprocessing

Pandas and NumPy libraries

Basic statistics and data analysis.

7. Project Objectives

Technical Objectives:

Develop a machine learning model capable of identifying fraudulent claims.

Performance Objectives:

Improve detection accuracy and reduce false positives.

Deployment Objectives:

Provide a simple user interface for claim input and prediction.

Learning Outcomes: Understand ML workflows including training, testing, and deployment.

8. System Workflow

Step 1: User opens the Streamlit application.

Step 2: User enters insurance claim details.

Step 3: Input data is processed and converted into numerical format.

Step 4: Trained machine learning model analyses the data.

Step 5: System predicts fraud probability.

Step 6: Result is displayed on the interface.

9. Requirement Analysis & System Design

Problem Definition:

Insurance companies process thousands of claims daily. Detecting fraudulent claims manually is difficult and time consuming.

Functional Requirements:

Accept claim inputs

Process claim information

Predict fraud

Display result

Non Functional Requirements:

Fast response

High reliability

Easy interface.

10. Environment Setup

Install Python and required libraries using pip.

Example command:

pip install pandas numpy scikit-learn streamlit joblib matplotlib

Create project files:

app.py

train.py

requirements.txt

11. Core System Development

Feature 1: Data Preprocessing

Cleaning dataset, handling missing values, and converting categorical variables.

Feature 2: Machine Learning Model

Training classification models such as Logistic Regression or Random Forest.

Feature 3: Streamlit Interface

Allows user input and displays prediction results.

12. Integration & Optimization

All components including dataset processing, model prediction, and user interface are integrated.

Performance improvements include optimized model parameters and efficient data preprocessing.

13. Testing & Validation

Testing ensures the system performs correctly.

Metrics used:

Accuracy

Precision

Recall

F1 Score

Example Test Case:

Input: Suspicious claim details

Expected Output: Fraud detected

14. Deployment

The system can be deployed locally or on cloud platforms such as:

Streamlit Cloud

Deployment Steps:

Train model

Save model using joblib

Run application using 'python -m streamlit run app.py'.

15. Project Structure

project_root/ app.py

train.py

requirements.txt

16. Results

The model predicts whether a claim is fraudulent or genuine based on the provided claim information.

Testing shows that machine learning can significantly help identify suspicious claims quickly.

17. Advantages & Limitations

Advantages:

Fast fraud detection

Reduces manual investigation

Improves efficiency

Limitations:

Dependent on dataset quality

May require periodic retraining

New fraud patterns may reduce accuracy.

18. Future Enhancements

Integration with real-time insurance databases

Deep learning based detection models

Mobile application interface

Cloud based deployment

Real-time fraud alert systems.

19. Conclusion

The project demonstrates the practical use of machine learning in detecting insurance fraud.

20. Appendix

A. Source Code Files

app.py – Streamlit application for user interaction and prediction.

train.py – Script used for training the machine learning model.

B. Configuration Files

requirements.txt containing the list of Python libraries required to run the project.

C. Dataset Details

Dataset name:

- employee_data.csv
- vendor_data.csv
- insurance_data.csv

Features include:

- Claim Amount
- Policy Holder Age
- Insurance Type
- Report Date
- Police Report

A1 AGENT_ID

	A	B	C	D	E	F	G	H	I	J	K
1	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID	AGENT_ID
2	AGENT000	Ray Johns	########	1402 Maggies Way		Waterbury	VT	5677	34584958	HKUN51252328472585	
3	AGENT000	Angelo Boı	########	414 Tanya Pass		Panama C	FL	32404	1.07E+08	OPIS19290040088204	
4	AGENT000	Candy Spe	########	606 Nation #306		Fayettevill	AR	72701	81744097	YSCJ67489688482590	
5	AGENT000	Mary Smitl	########	235 Hugh Thomas Dri		Panama C	FL	32404	67563771	ZANG21285355574581	
6	AGENT000	Mildred Di	########	3426 Broadview Stree		Montgome	AL	36110	1.15E+08	DZFS82244494451134	
7	AGENT000	Linda Butl	########	10383 Eagle River Lar		Anchorage	AK	99577	82970527	TIDG93550966796680	
8	AGENT000	Michael H	########	5011 West New Worl		Glendale	AZ	85302	53819552	OVFE89299102977291	
9	AGENT000	Carolyn Qu	########	2538 East 40th Plaza		Panama C	FL	32405	58311521	HOSW68784912234194	
10	AGENT000	Marilyn All	########	4840 Reservoir Road I		Washingto	DC	20007	66645522	NXRT72066780733033	

POLICY_EFF_DT	LOSS_DT	REPORT_DT	INSURANCE_TYPE	PREMIUM_AMOUNT	CLAIM_AMOUNT
23-06-2015	16-05-2020	21-05-2020	Health	157.13	9000
21-04-2018	13-05-2020	18-05-2020	Property	141.71	26000
03-10-2019	21-05-2020	26-05-2020	Property	157.24	13000
29-11-2016	14-05-2020	19-05-2020	Health	172.87	16000
26-12-2011	17-05-2020	22-05-2020	Travel	88.53	3000
28-12-2012	20-05-2020	25-05-2020	Life	87.02	63000
26-10-2012	13-05-2020	18-05-2020	Health	197.23	3000
30-12-2018	14-05-2020	19-05-2020	Motor	83.77	8000
27-06-2019	21-05-2020	26-05-2020	Motor	82.14	5000

D. Model Information

Algorithm Used: Random Forest / Logistic Regression

Training Method: Supervised learning classification

Evaluation Metrics: Accuracy, Precision, Recall, F1 Score

E. GitHub Repository Link

https://github.com/Tanvi7T/insurance-fraud-detection-ml

F. Live Demo Link

https://insurance-fraud-detection-ml-vmreirvrgqwigmkyczmzbe.streamlit.app/